

Verification & Validation / QA Ticket List

Implementation-ready specification for a Pomodoro productivity application inspired by the functional model of pomodorotimer.online. Revised scope: anonymous, local-first, no login/signup, no user authentication.

1. Validation Strategy

Verification proves the implementation meets the specification; validation proves the user workflow behaves correctly in realistic browser conditions. Release-blocking tests cover timer accuracy, persistence, local data integrity, offline behavior, privacy, accessibility, and production build health.

2. Feature Validation Matrix

Test	Area	Verification	Expected result
VAL-001	Timer initial	Load default route	Focus mode displays 25:00 by default.
VAL-002	Countdown	Run a short controlled timer	Displayed time follows wall clock within defined tolerance.
VAL-003	Background tab	Background timer for several minutes	Return shows correct remaining wall-clock time.
VAL-004	Refresh	Refresh while running	Running state and deadline restore correctly.
VAL-005	Close/reopen	Reopen before deadline	Remaining time is reconstructed; timer does not restart.
VAL-006	Expired reopen	Reopen after deadline	Session resolves once; no duplicate completion.
VAL-007	Pause	Pause then wait	Remaining time does not decrease while paused.
VAL-008	Resume	Resume paused timer	New deadline derives from saved remaining time.
VAL-009	Cycle	Complete four focus cycles	Configured long break is selected at correct interval.
VAL-010	Task CRUD	Create/edit/delete/complete tasks	State persists and reload matches latest state.
VAL-011	Task focus	Complete focus with selected task	Exactly one focus session is attributed to task.
VAL-012	Break accounting	Complete break on selected task	Task focus seconds do not increase.
VAL-013	Nested rollup	Track time on multiple children	Parent derived total equals descendant totals once.
VAL-014	Daily stats	Complete mixed sessions	Counts/duration match session records for local date.
VAL-015	Streak	Meet threshold on consecutive dates	Current/longest streak follow rules.
VAL-016	Freeze	Miss eligible day with freeze	Freeze applies once according to configured policy.
VAL-017	Challenge	Reach daily target	Progress reaches target once and reward unlocks.
VAL-018	Cross-tab start	Start in tab A	Tab B reflects running timer.
VAL-019	Cross-tab pause	Pause in tab B	Tab A reflects pause quickly.
VAL-020	Duplicate completion	Keep two tabs open through zero	Only one completed session/stat increment is stored.
VAL-021	Offline	Disable network after app cached	Timer/tasks/settings remain operational.
VAL-022	PWA	Run installability checks	Manifest/service worker/install criteria pass.
VAL-023	Notification deny	Deny permission	Timer remains fully usable.
VAL-024	PiP	Open supported PiP	Timer/control state stays synchronized.

Test	Area	Verification	Expected result
VAL-025	Custom image valid	Upload accepted image	Image is optimized, stored locally and selectable.
VAL-026	Custom image invalid	Upload invalid/oversized content	Rejected safely with understandable feedback.
VAL-027	Storage quota	Simulate quota failure	App handles failure without timer crash.
VAL-028	XSS	Enter HTML/script-like task content	Content renders as text; script does not execute.
VAL-029	Corrupt persistence	Inject malformed stored state	App validates/falls back instead of crashing.
VAL-030	Accessibility	Keyboard-only critical journey	All core controls reachable/operable with visible focus.
VAL-031	Reduced motion	Enable reduced motion	Non-essential motion is removed/reduced.
VAL-032	Mobile	Run critical flows at 320/375 widths	No blocked controls or unusable overflow.
VAL-033	Build	Run lint/typecheck/build	All commands exit successfully.
VAL-034	E2E	Run Playwright critical suite	All release-blocking journeys pass.
VAL-035	Security headers	Inspect production response	Configured CSP/security headers are present.
VAL-036	Privacy	Inspect network during personal actions	Task/session/background content is not sent to server.

3. Required Automated Layers

- Unit: timer math, state transitions, streak calculations, challenge progress, data validation.
- Integration: Zustand/services/IndexedDB repositories and migration behavior.
- E2E: timer lifecycle, refresh recovery, tasks, cross-tab where automation permits, offline/PWA smoke flows.
- Static gates: lint, TypeScript, production build, dependency/security review.

4. Release Gate

```
npm run lint npm run typecheck npm run test npm run build npm run test:e2e
```

5. Evidence Required

A feature must not be marked complete solely from an implementation claim. Store test output or CI evidence for automated checks and concise reproduction evidence for manual/browser-specific checks. Any failed release-blocking check returns the ticket to in-progress.

6. Browser Matrix

At minimum validate current stable Chromium, Firefox, and Safari/WebKit behavior where supported, plus Android Chrome and iOS Safari for core timer/task flows. Browser-specific features such as Picture-in-Picture may be conditionally available and must degrade gracefully.

7. Final Acceptance

- All P0 feature tickets complete.
- All release-blocking validation cases pass.
- No known duplicate-session or timer-drift defect.
- No login/signup requirement exists.

- Core personal data remains local by default.
- Production HTTPS/security headers verified.
- Mobile and offline smoke tests pass.